Disha Maheshwari

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EDUCATION

Purdue University Jan 2022 – May 2025

B.S. in Computer Engineering, GPA: 3.97/4.0 | *Minor in Mathematics* | *Certificate in Applications of Data Science*

Coursework: Data Structures, Operating Systems, Computer Networks, Computer Architecture, Artificial Intelligence, NLP, Software Engineering, Computer Graphics, ASIC Design, Computer Security

EXPERIENCE

Amazon | Software Development Engineer Jul 2025 – Apr 2026

- Developed and scaled distributed feature computation services for fraud detection handling 147M+ TPS in Java and 2.25M TPS in Python with 20–80 ms p99 latency.
- Led migration from HTTP to CloudAuth-enabled HTTPS, achieving 100% encryption-in-transit compliance for Tier-1 services.
- Deployed and scaled services across AWS regions for shadow testing, service migration, and high-traffic events; improved reliability and fault tolerance.
- Owned on-call operations including incident response, root cause analysis, and long-term reliability improvements for production systems.

Amazon | Software Development Engineer Intern May 2024 – Aug 2024

- Built and deployed a scalable authentication pipeline on AWS for a Java-based service.
- Designed S3-based data ingestion, in-memory caching, and periodic auth refresh workflows to improve performance and reduce latency.

Oak Ridge National Laboratory | Research Intern May 2023 – Aug 2023

- Implemented spiking neural networks on Artix-7 FPGA in Verilog to prototype neuromorphic hardware architectures.
- Designed UART, receiver, FIFO, and SNN modules and validated system functionality at 100 MHz on basic algorithmic workloads.
- Published a first-author IEEE paper on a low-cost, energy-efficient FPGA-based neuromorphic processor.

RESEARCH AND PROJECTS

Global Standard Deviation Subnet Inference (Bayesian Deep Learning) | Prof. Ruqi Zhang

- Developed SDSI, a scalable variational inference method for Bayesian neural networks using sparse low-dimensional variance subspaces.
- Improved uncertainty calibration and robustness on clean and corrupted datasets through a gradient-preserving subspace selection strategy.
- Achieved 76.36% accuracy, 0.9655 AUROC, and 0.0014 ECE on CIFAR-100, outperforming Sparse BNNs, VBLL, and Laplace Subnetworks.

Package Registry System (AWS, Distributed Systems)

- Built a scalable package registry with REST APIs for module upload, versioning, and retrieval using Node.js and AWS (EC2, S3, DynamoDB).
- Implemented dependency analysis, version tracking, and cost metrics to improve system reliability and performance.
- Developed CI/CD pipeline with GitHub Actions and deployed on AWS with automated testing and continuous deployment.
- Enhanced security using HTTPS, IAM, token-based authentication, and STRIDE threat modeling.

QueryForge | RAG-Based Research Discovery Platform

- Engineered QueryForge, a scalable RAG-powered platform to streamline discovery of academic research opportunities, reducing manual search across fragmented sources.
- Designed end-to-end pipeline: web scraping → embedding generation → vector search → LLM-based response generation, integrated into a Flask-based web app.
- Deployed using Docker + AWS ECS/ECR, ensuring reproducible and scalable production environments.

[Re] BiRT: Bio-Inspired Replay in Vision Transformers for Continual Learning

- Conducted a reproducibility study of BiRT, a continual learning framework using Vision Transformers and episodic memory replay to mitigate catastrophic forgetting.
- Evaluated benchmark performance, identified implementation gaps, and published findings as first author in ReScience C.

Wisebucks.AI | Founder

- Founded an ML-driven fintech platform and secured $6,000 in startup funding through the JMEC incubator competition.
- Built financial analytics and recommendation components using LSTMs, Monte Carlo simulation, sentiment analysis, SQL pipelines, APIs, and web scraping.
- Integrated chatbot support and portfolio insights using LLM-based workflows and financial data processing systems.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, SQL, JavaScript, HTML/CSS, Verilog
ML/AI: PyTorch, Scikit-learn, NLP, LLMs, RAG, Bayesian Deep Learning, Embeddings, Retrieval Systems
Cloud/DevOps: AWS (EC2, ECS, ECR, S3), Docker, CI/CD, Linux, Bash
Systems: Distributed Systems, Low-Latency Systems, Computer Architecture, Operating Systems
Tools: Git, Flask, Pinecone, Haystack, Pandas, NumPy, SciPy, Matplotlib, Vivado

PUBLICATIONS

An FPGA-based Neuromorphic Processor with All-To-All Connectivity, First Author, IEEE ICRC 2023 – doi.org/10.1109/ICRC60800.2023.10386808
[Re] Bio-Inspired Replay in Vision Transformers, First Author, ReScience C, 2025 – zenodo.org/records/14964875